

Comparing Clinical and Machine Learning Approaches for Cardiac Mortality Risk Prediction

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Abstract—Current mortality risk for patients with heart failure relies on clinical scores such as Meta-Analysis Global Group in Chronic Heart Failure (MAGGIC), which do not incorporate cardiac monitoring data or machine learning. In this study, we examine the performance of machine learning models on clinical, electrocardiographic, and laboratory features for prediction of cardiac mortality due to sudden cardiac death or pump-failure death. We rely on a publicly available dataset collected from the MUerte Subita en Insuficiencia Cardiaca (MUSIC) study cohort, consisting of 992 heart failure patients with clinical data, chest X-rays, 20-min high-resolution electrocardiograms (ECGs), and 24-hour Holter ECG monitoring. Five machine learning algorithms (XGBoost, CatBoost, Gradient Boosting, Random Forest, and Logistic Regression) were trained on 92 to 132 variables representing patient history, physical examination, laboratory testing, ECG parameters, and echocardiographic data, with class weighting explored across all models. Using 11 variables available in the dataset, we also calculated modified MAGGIC scores as a clinical baseline, producing an AUC score of 0.710. All machine learning models achieve an AUC score of at least 0.721, with class-weighted XGBoost demonstrating the highest AUC score of 0.777. The performance demonstrates risk prediction improvements from electrocardiographic and Holter monitoring data beyond traditional clinical variables. These findings support a complementary approach to clinical decision-making where traditional risk scoring and machine learning models enhance risk stratification for chronic heart failure patients.

Index Terms—Cardiac mortality, Heart failure, Sudden cardiac death, Machine learning

I. INTRODUCTION

Heart failure is a chronic condition where heart muscles progressively weaken and cannot pump enough blood to support the body [1]. It can result in a myriad of outcomes, such as sudden cardiac death (SCD) and pump failure death (PFD). SCD is defined as death from a cardiovascular cause occurring within one hour of symptom onset, an unwitnessed death in the absence of preexisting circulatory failure, or death during attempted resuscitation [2]. Identifying individuals with a high risk of SCD remains challenging [3]. For example, while coronary artery disease is the underlying cause of around 80% of SCD cases, appropriately 50% of these SCD events occur in patients with no prior diagnosis of any cardiovascular disease [4]. Meanwhile, PFD is defined as mortality from gradual deterioration of cardiac function and often occurs as end-stage heart failure [5]. One-third of newly diagnosed heart failure patients die within 3 years despite promising advancements in treatment [6]. Several medical studies collect

patient data to determine statistically relevant biomarkers that are most predictive of SCD and PFD.

One such longitudinal study is the MUerte Subita en Insuficiencia Cardiaca (MUSIC), which assesses risk predictors of cardiac death—SCD or PFD—for 992 patients with chronic heart failure (CHF) between April 2003 and December 2004 with a median follow up of 44 months [2]. MUSIC was specifically designed to develop and evaluate signal-derived biomarkers. In addition to demographics and lab values, a 20-min high-resolution ECG and 24-hour Holter ECG were collected for the majority of patients. High-resolution 3-lead ECGs capture subtle ventricular conduction abnormalities associated with SCD [7]. 24-hour Holter monitoring records arrhythmias, ventricular ectopic burden, and other cardiac electrical markers that help stratify risk in patients with heart failure [8]. MUSIC’s diverse inclusion of clinical, demographic, radiographic, echocardiographic, laboratory, and medication variables offers further insight on cardiac death onset compared to traditional risk predictors.

Traditional cardiac risk assessment is often based on a limited set of clinical parameters, such as the left ventricular ejection fractions (LVEFs) and systolic blood pressure. For instance, the Meta-Analysis Global Group in Chronic Heart Failure (MAGGIC) is a clinical tool that produces a 1- or 3-year mortality risk score for patients with heart failure [9]. This method is designed to rely on only 13 variables for ease of application by practitioners, but machine learning models can learn from a greater variety of clinical features, potentially improving prognostic accuracy [10].

The purpose of this paper is to evaluate five machine learning models trained on the clinical and signal-derived features of the MUSIC dataset and compare their risk performance to that of MAGGIC scores calculated from the same data. Our selected machine learning models are eXtreme Gradient Boosting (XGBoost), CatBoost, Gradient Boosting, Random Forest, and Logistic Regression, trained under both default and class-weighted settings. For MAGGIC, two metrics—recall and precision—are calculated across 6 risk-stratified thresholds and at an optimized threshold. Clinical model performance is assessed using area under the receiver operating curve (AUC) and area under the precision-recall curve (AUPRC). We use the same metrics for our machine learning models with optimized thresholds. To the best of our knowledge, this is the first study to derive MAGGIC risk scores from the MUSIC dataset.

The sections of this paper are as follows: Section II covers related works; Section III is a description of the data preprocessing; Section IV details experiment design with MAGGIC and the five machine learners; Section V contains results of the model performances; and Section VI concludes the paper with future areas of research.

II. RELATED WORK

The clinical risk score for heart failure patients selected for comparison to machine learning heart failure risk scores is MAGGIC. Pocock et al. use the meta-analysis data of 39,372 patients over 30 studies to assign a 1- and 3-year probability of death [9]. The range of follow-up period was 1.0–3.9 years with a median of 2.5 years. Multivariate piecewise Poisson regression methods with stepwise variable selection created a final model of 13 highly significant predictors of mortality, listed in descending order of impact: age, LVEFs, New York Heart Association (NYHA) class, serum creatinine, diabetes, not prescribed beta-blocker, lower systolic blood pressure (BP), lower body mass, time since diagnosis, current smoker, chronic obstructive pulmonary disease (COPD), gender, and not prescribed ACE-inhibitor or angiotensin-receptor blockers (ARB). Missing values were handled by 25-iterations of multiple imputations using chained equations (MICE) to create 25 imputed data sets. Model coefficients of the 25 data sets are averaged by Rubin’s rules to derive an integer risk score. Rubin’s rules are a set of formulas to combine parameter estimates from multiple imputed datasets. A pooled point estimate and total variance are integral to account for otherwise uncaptured trends in missing data. One goal of the study was to include patients with preserved LVEFs ($\geq 40\%$) because most previous models only used patients with reduced LVEFs ($< 40\%$). The results showed that MAGGIC risk scores closely followed the ground-truth values of 3-year patient mortality, but the predicted numbers for groups 4–6 were lacking compared to the observed values. In other words, patients who were at highest risk were not targeted for preventative care.

In 2014, Sartipy et al. conducted research to validate the MAGGIC risk scores [11]. The 3-year mortality risk for 51,043 patients from the Swedish Heart Failure Registry was calculated with overall success: observed mortality was 39.4% compared to predicted mortality at 36.4%. Missing values were also handled with 25-fold MICE methods. Eleven of the thirteen features needed for the score were readily available—the LVEF and heart failure duration variables needed re-categorization. Since the registry’s ejection fraction categories differed from MAGGIC’s requirements, researchers re-categorized patients by merging higher LVEF groups and randomly assigning hypothetical values to lower LVEF groups to match MAGGIC categories. Additionally, because heart failure duration was only recorded as above or below 6 months, all patients with duration under 6 months were classified as diagnosed within 18 months for the MAGGIC score calculation. The study focused on discrimination (C-index of 0.741) and calibration quality with the 3-year mortality risk strata.

In 2009, Vazquez et al. ran initial tests on the MUSIC cohort to develop prognostic risk scores for cardiac mortality, SCD, and PFD with CHF patients [5]. At the time, existing models resulted in extremely variable, generally poor outcomes. A multivariate Cox regression was used with the top 10 biomarkers: prior atherosclerotic vascular event, left atrial size $>26 \text{ mm/m}^2$, LVEF $\leq 35\%$, atrial fibrillation, left bundle-branch block or intraventricular conduction delay, non-sustained ventricular tachycardia and ventricular premature beats, estimated glomerular filtration rate $< 60 \text{ mL/min/1.73m}^2$, hyponatremia $\leq 138 \text{ mEq/L}$, NT-proBNP $> 1,000 \text{ ng/L}$, and troponin-positive. The authors calculated C-indices ranging from 0.74–0.8 for each type of cardiac mortality. The only variable that overlaps between the regression’s 10-feature set and the clinical risk score’s 13-feature set is LVEF, demonstrating how different variables representing patient information can improve risk prediction.

In 2017, Ramírez et al. combined ECG and clinical markers to predict SCD and PFD in risk models with MUSIC’s cohort [12]. Univariate Cox regressions were run to narrow down which variables were most predictive followed by multivariate Cox regressions to further refine which variables had a $p < 0.05$ with either SCD or PFD. A clinical model, ECG-based model, and combined model were run. The most impactful features for prognostic SCD predictions were gender, NYHA class, LVEF, T-wave alternans, T-peak-to-end restitution, and T-wave morphology restitution. PFD’s highest predictors were diabetes, NYHA class, T-peak-to-end restitution, and turbulence slope. The authors found that when predicting SCD, a set of ECG markers and clinical variables increased predictive power substantially, while for PFD, either the ECG features alone or combined set provide the best performance. This work demonstrates how the inclusion of variables beyond standard clinical attributes can strengthen prediction of cardiac death.

MAGGIC is a validated, widely used tool for predicting 1- and 3-year mortality for patients with heart failure, including preserved and reduced LVEFs. Nonetheless, direct, methodologically consistent comparisons of cardiac death risk scores derived from a traditional clinical test, such as MAGGIC, and from machine learning models remain limited. Vazquez et al. and Ramírez et al. demonstrate examples of statistical analyses and risk models built on the tabular data from the MUSIC database, but this application can be enhanced with machine learning [5], [12]. This work integrates machine learning into clinical decision making by evaluating the performance of five classifiers against a prognostic baseline.

III. DATA PREPROCESSING

The study population of the MUSIC cohort consists of 992 CHF patients enrolled in specialized heart failure clinics in eight University Spanish Hospitals between April 2003 and December 2004 with follow-up medians of 44 months, up to November 2008 [2]. The majority of patients had 3-lead, high-resolution resting ECGs, 24-hour 2- or 3-lead Holter ECGs, chest X-rays, echocardiographs, and blood laboratory tests performed upon study enrollment. Primary outcomes at the

end of the follow up were cardiac death—either SCD or PFD. The database was made publicly available on PhysioNet in September 2024. We used Version 1.0.1, which removes two redundant attributes and corrects ECG header files and lead count for high-resolution ECGs [13].

MUSIC has 936 instances of Holter ECGs and 687 instances of high-resolution ECGs. Each recording contains an ECG signal with corresponding header information about the recording format. Standard ECG intervals and morphology were calculated from the files by the MUSIC creators: PR interval, QRS duration, QT interval (with rate correction), Q-wave presence, left ventricular hypertrophy criteria, and rhythm classification. The Holter-derived metrics include RR interval statistics (minimum, maximum, average, and range), arrhythmia burden (ventricular and supraventricular premature beat counts, and tachycardia episodes), and time-domain heart rate variability indices (SDNN, SDANN, RMSSD, and pNN50). The MUSIC dataset also contains two features extracted from the chest X-rays: signs of pulmonary venous hypertension and cardiothoracic ratio.

The subject info dataset contains 992 instances and 103 features describing clinical tests, demographics, laboratory tests, medication, radiographs, and echocardiographs. Of these, 43 were categorical variables (including binary yes/no variables and ordinal/nominal variables with 1-9 categories), while 58 were continuous or discrete numeric measurements. The two remaining features consisted of a temporal variable and a unique patient identifier.

As a preprocessing step, unnecessary or redundant features were removed from the MUSIC dataset. We removed the Patient ID and the target variable. Then, any study logistic attributes were omitted, including the follow-up period length features, the exit of the study, the availability indicators for the high-resolution and Holter ECG data, and the Holter onset time. Lastly, three features that were or could be derived from other features were removed: body mass index (BMI), QRS > 120 ms, and the RR range.

The target variable “Cause of death” originally consisted of four classes: survival, non-cardiac death, SCD, and PFD. Since our study is focused on cardiac mortality, we dropped 61 instances of non-cardiac death and binary encoded the target (0: survival; 1: cardiac death). The target class distribution is 726:205, reflecting a class imbalance of approximately 3.54:1, with cardiac death as the minority. The remaining preprocessing steps were tailored separately for the MAGGIC score, for CatBoost and XGBoost, and for the remaining machine learning models.

A. MAGGIC Clinical Risk Score

Following established clinical prediction model guidelines, 13 variables are required for MAGGIC score calculation. However, two binary variables are not directly available in the dataset—COPD diagnosis and heart failure diagnosed within 18 months—and are therefore omitted. Consequently, we calculate a modified MAGGIC risk score using 11 variables without reweighting. The predicted risk is biased slightly

downward, for those two features would add 2 points each to the integer risk score if true.

Some of the variables needed to be preprocessed to align with the risk calculation standards. Creatinine was converted from $\mu\text{mol/L}$ to mg/dL by dividing the values by 88.4. Derived variables were created including current smoker status (from daily cigarettes smoked) and combined ACE inhibitor/ARB medication use. All necessary features were numerical (binary, continuous, or ordinal), requiring no categorical encoding. Missing data were handled using MICE following the methodology of Sartipy et al. [11]. This created 25 multiply-imputed datasets. MAGGIC scores were calculated for each of the 25 imputed datasets and pooled using Rubin’s rules to obtain final risk scores with associated standard errors. The finalized clinical dataset retained 11 predictor variables for risk score calculation: age, gender, BMI, systolic blood pressure, creatinine, diabetes, current smoker, beta-blocker, ACE-I or ARB, LVEF (%), and NYHA class.

B. Machine Learning Models

CatBoost and XGBoost are both gradient-boosting decision tree methods that have native categorical and missing value handling [14]. As such, we did not impute missing numerical values and only converted our categorical features to an object data type for CatBoost or to a category data type for XGBoost. Missing values in categorical features were treated as a distinct category by CatBoost’s ordered boosting algorithm [15]. Meanwhile, XGBoost uses a sparsity-aware technique that learns the optimal direction for missing values during the split finding process when building decision trees [16]. This approach results in 92 features in the dataset.

For models without native categorical or missing value handling (Gradient Boosting, Random Forest, and Logistic Regression), data preprocessing is split between the numerical and categorical features. For missing numerical values, we performed mean imputation and generated 40 binary missing indicator columns to preserve potentially informative missingness patterns. For missing categorical values, we imputed the string “Unknown”. Then, target-based encoding is applied to the categorical features using the CatBoost encoder, generating target-informed continuous representations [17].

IV. EXPERIMENTAL DESIGN

After preprocessing, we perform a stratified 80:20 split to produce a training set of 744 instances and test set of 187 instances. All models, including MAGGIC, are evaluated on the same held-out data for consistency. There are several clinical tests traditionally used to calculate risk of death from CHF. MAGGIC serves as the baseline model since it is an established benchmark. It uses clinical and lab data, no high-resolution ECG or Holter ECG features. Although both 1- and 3- year risk scores for heart failure mortality are derived, Pocock et al. and Sartipy et al. validated their results solely with the 3-year score; we follow this procedure in our own evaluation [9], [11].

We transform the integer risk scores to predicted probabilities by using the conversion table provided in Pocock et al. to produce the AUC and AUPRC values [9]. Binary predictions were derived using two approaches. The first follows the traditional procedure of six risk-stratified score groups. We set the lower cutoff as the decision threshold and measure the performance at each threshold with recall and precision. Second, we use an optimized threshold based on the training data to facilitate direct comparison of performance against the machine learning models.

To create the baseline model performance with machine learners, five selected models were run using default hyperparameters at an optimized decision threshold based on 5-fold cross-validation. XGBoost, CatBoost, Gradient Boosting, and Random Forest were selected since tree-based methods are historically strong on tabular classification tasks [18]. Gradient Boosting is foundational to the historical development of clinical cardiac risk prediction. The model sequentially builds an ensemble of decision trees, where each tree corrects errors made by previous trees [19]. As the foundational boosting algorithm, it serves as an important baseline for comparison against its modern successors—XGBoost and CatBoost—which introduce algorithmic optimizations. XGBoost has built-in regularization to prevent overfitting and is capable of modeling non-linear relationships, which is ideal on complex cardiovascular prediction tasks [20]. CatBoost handles imbalanced data through automatic class weighting and uses boosting to reduce overfitting [14]. Random Forest is adept at handling overfitting through averages of multiple decision trees. It handles high-dimensional medical data without extensive feature selection or engineering [21]. We also included Logistic Regression since the model is common in traditional medical and statistical analysis. It uses coefficients that are transformed into odds ratios, making them comparable to clinical cardiac risk tests [22]. Feature scaling was only applied for Logistic Regression, where numeric features are standardized by removing the mean and scaling to unit variance based on the training data.

We ran an additional configuration with four models using class weighting, where weights are assigned inversely proportional to class frequencies. For CatBoost, Logistic Regression, and Random Forest, their respective class weight parameters was set to "balanced". For XGBoost, the positive class weight was set to 3.54 to reflect the observed class imbalance. Since Gradient Boosting does not have a direct class weight parameter, we only evaluated the model with its default settings.

Our performance metrics are largely derived from true positive (TP), true negative (TN), false positive (FP) and false negative (FN) values. Precision is defined as the proportion of predicted positives that are observed positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

Recall is the proportion of observed positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

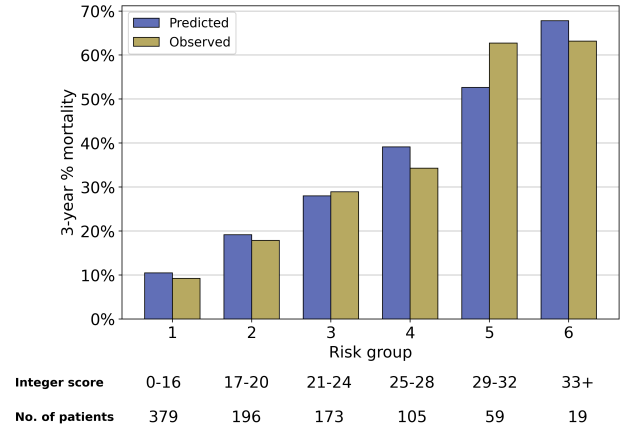


Fig. 1. Observed vs. MAGGIC-predicted 3-year mortality in six risk groups for full dataset.

Precision and recall are threshold-dependent metrics, as they rely on a fixed classification cutoff, whereas AUC and AUPRC are threshold-independent metrics. AUC measures a model's ability to discriminate between classes across all classification thresholds on a scale of 0 to 1. It is defined as the area under the curve of the true positive rate (TPR) versus the false positive rate (FPR). Meanwhile, AUPRC is defined as the area under the precision-recall curve, capturing the trade-off between precision and recall over all classification thresholds. This metric is particularly informative for imbalanced datasets, such as the MUSIC dataset [23].

Rather than using a fixed threshold across the models, we optimized classification thresholds by maximizing the F1-score on out-of-fold predictions from 5-fold cross-validation on the training set. The selected threshold was then applied without modification to the held-out test set, preventing the data leakage. The F1-score, defined as

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

represents the harmonic mean of precision and recall. This approach provides a balanced measure of classification performance for imbalanced datasets and enables fair comparison across models.

V. RESULTS

A. MAGGIC Risk Score

The traditional risk score derived from MAGGIC corresponds to a 3-year mortality risk. The distribution of MAGGIC scores across the full cohort is shown in Figure 1 to illustrate the overall risk stratification. However, all performance evaluation is restricted to the held-out test set to ensure consistency with the machine learning models. Table I presents the performance of MAGGIC across the six risk groups.

In Figure 2, MAGGIC both overestimates and underestimates the 3-year mortality across the risk groups, with performance varying by score range. In risk groups 1–4 (scores

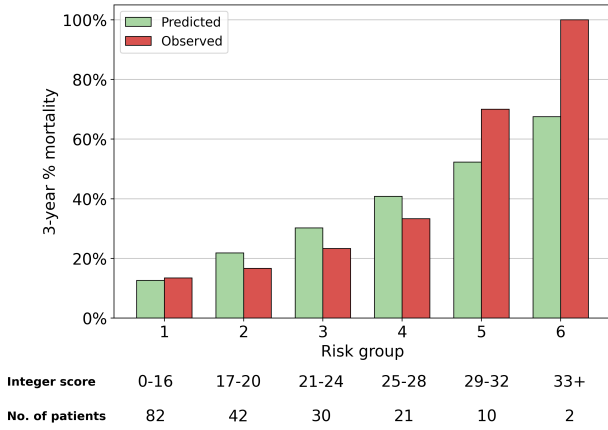


Fig. 2. Observed vs. MAGGIC-predicted 3-year mortality in six risk groups for test set.

0–28), predicted and observed mortality are relatively well-aligned with differences ranging from 0.8% to 7.5%. This suggests the model has adequate calibration for the majority of patients. However, the higher risk groups (scores 29–32 and 33+) reverses the trend of overestimation, as MAGGIC underestimates mortality by 17.7% to 32.5%, indicating reduced accuracy in identifying high-risk patients. Notably, these two risk groups have a combined sample size of 12 patients, limiting the estimation reliability. For comparison, Figure 1 demonstrates that MAGGIC produces more robust calibration for the two risk groups on a wider cohort. Overall, while MAGGIC provides reasonable risk stratification at the lower score ranges, it has difficulty predicting the patient population with elevated cardiac death risk, but this instability is likely driven by the limited sample size and two missing variables.

TABLE I
PERFORMANCE ACROSS MAGGIC RISK SCORE THRESHOLDS

Threshold	Recall	Precision	AUC	AUPRC
0	1.000	0.219		
17	0.732	0.286		
21	0.561	0.365	0.710	0.485
25	0.390	0.485		
29	0.220	0.750		
33	0.049	1.000		

Despite these challenges, MAGGIC achieved an overall AUC of 0.710 with the test set, indicating moderate discriminative ability, consistent with performance reported in prior heart failure cohorts. For example, Sartipy et al. reported a C-index of 0.741 across a substantially larger cohort of 51,043 patients, which is a closely related measure of discrimination that is broadly comparable to AUC [11]. Our AUC score translates to a 71% probability that a randomly selected patient that experienced a cardiac death is assigned a higher risk score than a randomly selected survivor. The mean MAGGIC score was 17.95 for the 187 patients, with values ranging from 3.0 to 35.0. Optimal performance occurred at the score

threshold of 21, which offers a relatively balanced recall and precision of 56.1% and 36.5%, respectively. This threshold could be a strong benchmark for ruling patients out of high-risk classification or for enrolling them in future heart health screenings. At the score threshold of 33, the precision of 100% indicates that all very high-risk patients within the MUSIC test set experienced cardiac death. These patients should be prioritized for immediate clinical intervention.

B. Machine Learning Models

Table II contains the performance of MAGGIC and the five machine learning models at their optimal thresholds with the best and worst results per metric in bold and italics, respectively. Due to space constraints, only the best-performing configuration for each model is reported; CatBoost and XGBoost use class weighting, while Gradient Boosting, Random Forest, and Logistic Regression are shown using their default settings and denoted as GB, RF, and LR, respectively. For MAGGIC, the F1-optimized probability threshold of 0.250 closely aligns with the score cutoff of 21, yielding identical classification performance.

All five algorithms achieved AUC values exceeding 0.72, outperforming the MAGGIC risk score. However, the recall scores of the machine learning models are subpar compared to MAGGIC, except for CatBoost, which matched MAGGIC, and Random Forest which produced the highest recall. Using the 92-feature dataset and class weighting, XGBoost and CatBoost achieved the highest discriminative performance with AUC scores of 0.777 and 0.763, respectively. In addition, XGBoost produced a higher AUPRC of 0.530 compared to 0.503 for CatBoost, which were the second- and third-highest AUPRC scores overall. This performance represents tangible improvements compared to the MAGGIC baseline across most metrics, with recall being largely comparable.

TABLE II
PERFORMANCE OF MAGGIC AND MACHINE LEARNING MODELS BASED ON RECALL, PRECISION, AUC, AND AUPRC

Model	Threshold	Recall	Precision	AUC	AUPRC
MAGGIC	0.250	0.561	0.365	0.710	0.485
CatBoost	0.325	0.561	0.434	0.763	0.503
XGBoost	0.150	0.439	0.486	0.777	0.530
GB	0.213	0.537	0.440	0.743	0.534
RF	0.246	0.659	0.386	0.753	0.488
LR	0.260	0.512	0.525	0.721	0.492

Trained on the 132-feature dataset, the Gradient Boosting model produced the highest AUPRC score of 0.534 out of all the models. Notably, Random Forest yielded the highest recall of 0.659 at the expense of having the second-lowest precision at 0.386. In addition, Random Forest achieved the second-lowest AUPRC of 0.488, only slightly above that of MAGGIC. Compared to the other four learners, Logistic Regression exhibited reduced overall discrimination based on AUC, but the best balance between precision and recall, along with the highest precision.

To determine the machine learning model with the best discriminative ability, we must consider both the threshold-independent and threshold-dependent metrics. XGBoost demonstrated the highest AUC, indicating relatively better discrimination compared to the other models. This performance was achieved using the 92-feature dataset without categorical encoding and missing value imputation, leveraging its built-in handling of these processes. This represents a practical strength, as this algorithmic design reduces pre-processing complexity while maintaining strong predictive performance. In contrast, Gradient Boosting achieved the highest AUPRC but required the fully imputed and encoded 132-feature dataset. These results suggest that while Gradient Boosting may better balance precision and recall at an optimized threshold, XGBoost provides the most robust overall discrimination.

VI. CONCLUSION

In this study, we applied the publicly available MUSIC dataset to predict cardiac mortality using MAGGIC as a baseline clinical tool and five machine learning models. The MAGGIC risk score draws on substantial clinical research and validation across diverse populations with large cohorts. Our findings conclude that machine learning offers superior detection of relationships among features, underscoring the value of cardiac monitoring data. In particular, class-weighted XGBoost achieved the highest performance based on AUC due to its ability to natively handle missing values and categorical features. Features derived from electrocardiography captured temporal dynamics, arrhythmic burden, and subtle electrical abnormalities that are not reflected in the static clinical measurements used in the MAGGIC score. The models' ability to integrate MUSIC's high-dimensional, heterogeneous physiological measurements, diagnostic codes, and temporal monitoring data demonstrates machine learning's capacity to identify complex risk patterns that may be difficult to capture with traditional clinical risk tools.

Future work could incorporate additional datasets and clinical tests, such as the Seattle Heart Failure Model, for comparison [24]. Regarding the signal-derived feature subset, this research used expert-extracted features from ECG and imaging as opposed to raw waveforms. Future work could explore whether deep learning on raw signals captures information beyond the expert-derived features.

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